

Patrick E. Carlson

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Education

Iowa State University	Ames, Iowa
PhD, Cumulative GPA: 3.68/4.0	2008 - 2015
Major: Human-Computer Interaction (HCI)	
Simpson College	Indianola, Iowa
Bachelor of Arts, cum laude, Cumulative GPA: 3.715/4.0	2003 - 2008
Majors: Computer Science and Psychology	

Experience

Sandia National Labs	Albuquerque, New Mexico
Data Scientist (promoted to Principal level - May 2021)	2017 - present
Data analysis, R and Python programming, data virtualization, Tableau and Shiny visualizations, analytics team co-lead, SQL and Mongo querying, architecture and data engineering research, project planning, writeups, design mockups, and UX reviews.	
Renaissance Learning	Wisconsin Rapids, Wisconsin
User Experience Analyst	2016 - 2017
Perform remote usability testing on software prototypes, help with UI design, and develop and analyze survey data.	
Iowa State University	Ames, Iowa
Improving Open Source Software Development (Dissertation Topic)	2010 - 2015
Apply data mining in the development of algorithms and tools to improve developer understanding of technical and social structure in large Open Source communities. Under direction of Dr. Judy Vance.	
Bimanual Haptics for Virtual Assembly Tasks (Research Assistant)	2009 - 2015
Designed and analyzed studies examining various bimanual haptic device configurations and learning transfer for virtual assembly tasks. Under direction of Dr. Judy Vance.	
Mentor/Support Research Experience for Undergraduate (REU) Students	Summer 2010
Helped coordinate three undergraduate REU students whose project entailed creating a physical mockup of a shopping cart which was then used in a virtual reality user study. Under direction of Dr. Judy Vance.	
University of New Mexico	Albuquerque, New Mexico
Research Experience for Undergraduates (REU)	Spring 2006 - Fall 2007
Learned the basics of machine learning and worked on applying this to improving the routing of packets in a dynamic network. Under direction of Dr. Terran Lane.	
Simpson College	Indianola, Iowa
Usability and Appeal of the Linux Desktop (Psychology Capstone)	Spring 2007 - Fall 2008
Constructed and performed an experiment empirically contrasting usability and appeal of a 2D versus 3D Linux desktop.	

Simulated the evolution of cooperation/altruism using game theory techniques and genetic algorithms.

Presented work at Argonne National Laboratory.

Teaching and Other

Engineering/LAS Online Learning (Graduate Assistant)	Fall 2014
ME/WLC 484/584: Technology, Globalization, and Culture (Teaching Assistant)	Fall 2013
HCI/Psych 522: Scientific Methods in HCI (Teaching Assistant)	Spring 2013
CS 309: Software Development Practices (Teaching Assistant)	Spring 2009
CS 207: Programming I (Teaching Assistant)	Fall 2008

Groups and Activities

Sandia Data Sciences - Community of Practice Organizer	2018 - Present
Iowa State University Badminton Club	2009 - 2015
ISU Human-Computer Interaction Student Group Vice President	2010 - 2011
Simpson College Math Club	2006 - 2008
Simpson College Computer Club	2003 - 2008

Honors

Nominated for Employee Recognition Award (ERA)	March 2026
Nominated for Lab Director's Award - Sandia AI Chat Team	May 2025
Nominated for Employee Recognition Award (ERA) - Hybrid Work Environment	March 2025
Awarded Employee Recognition Award (ERA) - Sandia AI Chat Team	March 2025
Awarded Employee Recognition Award (ERA) - Pioneering Machine Learning and Data Analytics Techniques to Mitigate Lab-wide Risk	March 2025

Recent Poster Presentations

Carlson, Patrick & Vance, Judy M. (2013, April). [Who Should I Contact?: Helping New Developers Find Experts](#). Poster presented at the Emerging Technologies Conference, Ames, IA. **Awarded most interesting research project.**

Peters, Anicia & Carlson, Patrick & Gilbert, Stephen & Vance, Judy M. (2012, April). [A Hybrid Method to Support Natural Interaction of Parts in a Virtual Environment](#). Poster presented at the Emerging Technologies Conference, Ames, IA. **Awarded most interesting research project.**

Recent Publications

Bruno, Sean & Carlson, Patrick (2023). Podman vs Docker comparison via R Shiny project deployment. In [Proceedings of the 2023 Improving Scientific Software Conference](#), pages 10-14.

Carlson, Patrick (2015). Engaging developers in open source software projects: harnessing social and technical data mining to improve software development. In [Graduate Theses and Dissertations](#), pages 1-192, Ames, IA.